

The POSIX Lexer Cubic Bound: the Set-Ledger Gate (Open Problem)

POSIX backref/cubic project — secretary session
repository github.com/hellotommy/posix, branch `codex/backref-values`

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Abstract

POSIX lexing by Brzozowski/Antimirov derivatives stays fast only if the simplified derivative states stay small. This note gives a fully self-contained statement of the one inequality this project still has to prove: after one step of the strongly simplified Antimirov-style row transition, the total size of the *deduplicated* set of opened linear forms is cubic in the size of the original regular expression. All definitions below are faithful translations of the checked Isabelle/HOL definitions in the repository (exact names in the table of Section 8); every fact labelled “checked” is proved in Isabelle with no **sorry**. Section 5 records what is already proved, Section 6 what is *refuted* and must not be attempted again.

1 Regular expressions and size

Regular expressions over an alphabet Σ (Isabelle skeleton type `rrex`p, restricted to its classical—“legacy”—fragment¹):

$$r ::= \mathbf{0} \mid \mathbf{1} \mid c \mid r_1 \cdot r_2 \mid \Sigma[r_1, \dots, r_n] \mid r^* \mid r^{\{n\}} \quad (c \in \Sigma, n \in \mathbb{N})$$

where $\Sigma[\cdot]$ is a *list* alternative (ordered, possibly with duplicates) and $r^{\{n\}}$ is bounded repetition (RNTIMES). Size $|\cdot|$ and list size $\|\cdot\|$ (`rsize`, `rsizes`):

$$|\mathbf{0}| = |\mathbf{1}| = |c| = 1, \quad |r_1 \cdot r_2| = 1 + |r_1| + |r_2|, \quad |\Sigma rs| = 1 + \sum_{r \in rs} |r|,$$

$$|r^*| = 1 + |r|, \quad |r^{\{n\}}| = 1 + |r| + n, \quad \|rs\| = \sum_{r \in rs} |r| \quad (rs \text{ a list}).$$

Note the repetition counter n itself counts toward size. Nullability ν (`rnullable`) is standard; $\nu(r^{\{0\}})$ holds.

2 The transition pipeline

List helpers. `flts` (`rflts`) deletes $\mathbf{0}$ members and splices one level of nested alternatives: `flts(0#rs) = flts rs`; `flts((Σts)#rs) = ts ++ flts rs`; otherwise the head is kept. `nub` (`rdistinct`, with empty accumulator) keeps the first occurrence of each member, preserving order. The smart alternative constructor $\oplus$ (`rsimp.ALTs`) maps $[] \mapsto \mathbf{0}$, $[r] \mapsto r$, and longer lists $rs \mapsto \Sigma rs$.

¹The formal skeleton carries three additional proof-only backreference constructors (`RBACKREF4`, `RHALF`, `RRESIDUE`). They have no partial derivatives and are excluded here by the predicate `legacy_rrex`; the open problem lives entirely in the classical fragment.

Sequence normalizer. σ (rsimp4.SEQ_atom) builds a right-associated sequence while erasing units:

$$\sigma(\mathbf{0}, t) = \mathbf{0}, \quad \sigma(\mathbf{1}, t) = t, \quad \sigma(p \cdot q, t) = \sigma(p, \sigma(q, t)),$$

and for an atom a (none of $\mathbf{0}, \mathbf{1}, \cdot$): $\sigma(a, \mathbf{0}) = \mathbf{0}$, $\sigma(a, \mathbf{1}) = a$, otherwise $\sigma(a, t) = a \cdot t$. On top of it, σ^* (rsimp7.SEQ_atom) absorbs adjacent equal stars:

$$\sigma^*(p^*, p^*) = p^*, \quad \sigma^*(p^*, p^* \cdot k) = p^* \cdot k, \quad \sigma^*(p, t) = \sigma(p, t) \text{ otherwise.}$$

Partial derivatives. The Antimirov partial-derivative list pder_c (rpder_list):

$$\begin{aligned} \text{pder}_c(\mathbf{0}) &= \text{pder}_c(\mathbf{1}) = [], & \text{pder}_c(d) &= \text{if } c=d \text{ then } [\mathbf{1}] \text{ else } [], \\ \text{pder}_c(\Sigma rs) &= \text{concat}(\text{map } \text{pder}_c \text{ } rs), \\ \text{pder}_c(p \cdot q) &= \text{map } (\lambda x. \sigma(x, q)) (\text{pder}_c p) \mathbin{++} (\text{if } \nu(p) \text{ then } \text{pder}_c q \text{ else } []), \\ \text{pder}_c(p^*) &= \text{map } (\lambda x. \sigma(x, p^*)) (\text{pder}_c p), \\ \text{pder}_c(p^{\{n\}}) &= \text{if } n=0 \text{ then } [] \text{ else } \text{map } (\lambda x. \sigma(x, p^{\{n-1\}})) (\text{pder}_c p), \end{aligned}$$

and its normalized form $\text{npder}_c(r) = \text{map } (\lambda x. \sigma(x, \mathbf{1})) (\text{pder}_c r)$ (rpder_norm_list).

Multi-step factored states. One factored step on a row list (afactored_step = rpder_norm_rows) and its iteration (afactored1):

$$\text{step}_c(rs) = \text{nub}(\text{flts}(\text{concat}(\text{map } \text{npder}_c \text{ } rs))), \quad F(r, \varepsilon) = [r], \quad F(r, s \ c) = \text{step}_c(F(r, s)).$$

Note: *no* strong pruning happens inside F ; the strong simplifier enters only in the single step that the open problem measures.

Strong simplifier. The pairwise shared-suffix pruner (rsimpStrong_prune_pair_row): an earlier row e prunes a later row l by

$$P(e, l) = \begin{cases} \sigma^*(\oplus(ms \setminus ls), k) & \text{if } e = (\Sigma ls) \cdot k \text{ and } l = (\Sigma ms) \cdot k, \\ l & \text{otherwise,} \end{cases}$$

where $ms \setminus ls$ removes from the list ms every member occurring in ls (rprune_eq_against), keeping order. The scan (rsimpStrong_prune_rows_raw) folds P over the already emitted rows:

$$\text{scan}(\text{seen}, []) = [], \quad \text{scan}(\text{seen}, r \# rs) = r' \# \text{scan}(r' \# \text{seen}, rs),$$

where $r' = (\text{fold}_{e \in \text{seen}} P(e, \cdot))(r)$, and $\text{prune}(rs) = \text{scan}([], rs)$. The strong simplifier S (rsimpStrong_raw):

$$\begin{aligned} S(\mathbf{0}) &= \mathbf{0}, \quad S(\mathbf{1}) = \mathbf{1}, \quad S(c) = c, \quad S(p \cdot q) = \sigma^*(S(p), S(q)), \\ S(\Sigma rs) &= \oplus(\text{nub}(\text{flts}(\text{prune}(\text{flts}(\text{map } S \text{ } rs))))), \quad S(p^{\{n\}}) = S(p)^{\{n\}}, \\ S(p^*) &= \begin{cases} \mathbf{1} & \text{if } S(p) \in \{\mathbf{0}, \mathbf{1}\}, \\ q^* & \text{if } S(p) = q^*, \\ S(p)^* & \text{otherwise.} \end{cases} \end{aligned}$$

The strong one-step row transition. (rpder_strong_rows_raw; this is the object the problem measures.)

$$\Delta_c(rs) = \text{nub}(\text{flts}(\text{prune}(\text{flts}(\text{concat}(\text{map } (\text{map } S \circ \text{npder}_c) \text{ } rs))))).$$

We also name the *generated* rows before simplification: $\text{gen}_c(rs) = \text{concat}(\text{map } \text{npder}_c \text{ } rs)$.

3 The ledger objects

The *opener* D (row_dlforms) unfolds a row into its deduplicated linear forms²:

$$D(\mathbf{0}) = \emptyset, \quad D(\Sigma rs) = \bigcup_{q \in rs} D(q), \quad D((\Sigma ps) \cdot k) = \bigcup_{p \in ps} D(\sigma^*(p, k)),$$

and $D(q) = \{q\}$ for every other shape. For a row list, $D(rs) = \bigcup_{q \in rs} D(q)$ (row_dlformss). The *set ledger* of a finite set U of regexes (rsize_set):

$$\|U\| = \sum_{q \in U} |q|.$$

4 The open problem

Conjecture 1 (set-ledger cubic gate). *For every classical (legacy) regular expression r , every string $s \in \Sigma^*$ and every character $c \in \Sigma$:*

$$\| D(\Delta_c(F(r, s))) \| \leq 2(|r| + 3)^3.$$

In words: run the normalized Antimirov row automaton from r over s , apply *one* strongly simplified step for c , open every resulting row into linear forms, deduplicate *across the whole union*, and sum the sizes of the distinct opened rows. The claim is that this total is cubic in the size of the original expression — uniformly in s , whose length does not appear on the right-hand side.

Remark 1. The constant $2(\cdot + 3)^3$ is the project’s working target; a checked bound $K(|r| + 3)^3$ for any absolute constant K , or for the $r^{\{n\}}$ -free sub-fragment, is already valuable (see Section 7).

Remark 2. The inequality is stated for arbitrary r in the consuming contracts; the checked supporting facts that feed it (head linearity, the discharged gate instance below) carry the hypothesis `legacy_rrexp  $r$` , so Conjecture 1 should be read with that hypothesis. Empirically the left-hand side grows only *quadratically* on exhaustive and random probes (checkpoint 2026-06-12, “multi-step union ledger probe”).

Remark 3 (why this matters). The conjecture is exactly the missing *size half* of the per-step state budget for the strongly simplified POSIX lexer. The repository already contains checked contracts (`rpder_strong_dcanon_rows_raw_afactored1_actual_dlforms_tight_cubic_contractl`, ... `_budgetsl`) that take Conjecture 1 as their only numeric assumption and return the full canonical-row contract: language correctness of the canonical rows together with cubic bounds on their length, cardinality, linear-term count, and total size. The value/POSIX-correctness side of the lexer is independently checked and waiting (`strong_deferred_memo_lexer_eq_lexer` etc.).

5 What is already checked

All of the following are Isabelle-checked, unconditional unless stated.

1. *Per-row bounds.* $\#D(q) \leq |q|$ (card_row_dlforms_le_rsize)
 $\|D(q)\| \leq |q|^2$ (rsize_set_row_dlforms_le_rsize_sq)
2. *Union bridges.* $\|D(rs)\| \leq \sum_{q \in rs} |q|^2$, instantiated to the actual one-step output (rsize_set_row_dlformss..._le_sum_rsize_sq)

²Recursion terminates by the size measure: when $p \in ps$, $|\sigma^*(p, k)| < |(\Sigma ps) \cdot k|$ (rsize_rsimp7_SEQ_atom_member_lt_RSEQ_RALTS).

3. *Card half, one degree.* $\#D(\Delta_c(rs)) \leq \|\text{gen}_c(rs)\|$
(card_row_dlformss_rpder_strong_rows_raw_le_generated)
4. *Deleter chain.* $\|\Delta_c(rs)\| \leq \|\text{gen}_c(rs)\|$, and the same budget bounds length, cardinality and linear-term count of $\Delta_c(rs)$
(rsizes_rpder_strong_rows_raw_le, rpder_strong_rows_raw_generated_budget)
5. *Head linearity (legacy).* every sequence head h with $h \cdot t \in D(\Delta_c(F(r, s)))$ satisfies $|h| \leq 1 + 2|r|$
(actual_union_seq_head_size_linear)
6. *Discharged gate instance (legacy; 2026-06-12).* With $H = 1 + 2|r|$ and $M = 1 + H + \|\text{gen}_c(F(r, s))\|$:

$$\begin{aligned} \|D(\Delta_c(F(r, s)))\| &\leq \underbrace{\|\text{nonseq}(U)\|}_{(1)} + \underbrace{\#\text{fronts}(r, sc) \cdot \sum_{t \in \text{tails}(U)} (1 + H + |t|)}_{(2)} \\ &\quad + \underbrace{\text{pairbudget}(r, s, c) \cdot M}_{(3)} \end{aligned}$$

where $U = D(\Delta_c(F(r, s)))$, $\text{nonseq}(U)$ are its non-sequence members, $\text{tails}(U)$ the set of tails of its sequence members, $\text{fronts}(r, u)$ a syntactic carrier of front terms ($\text{strong_derivative_front_terms}$, members linear by item 5), and pairbudget the shared-suffix pair budget of the one-step dlform universe ($\text{raw_shared_prune_active_suffix_pair_budget}$ over $\text{afactored1_strong_dlform_universe}$)

(actual_union_pair_budget_gate_instance)

7. *Summand (1) is cubic.*
(rsize_set_rnonseq_members..._afactored1_cubic)
8. *Fragment result.* If r is $r^{\{n\}}$ -free (and normal-form), then $\|F(r, s)\| \leq 2(2|r| + 3)^3$
(rsizes_afactored1_rntimes_free_rsize_cubic)
9. *Summand (3) vanishes (2026-06-12, 20:44).* The dlform universe contains only opened rows, and opened sequence heads are never alternatives (item 6's carrier), so no universe member has the keyed shape $(\Sigma \text{rows}) \cdot k$; hence the active-suffix key set is empty and $\text{pairbudget}(r, s, c) = 0$ identically. The gate instance has exactly two summands.
(pair_budget_afactored1_strong_dlform_universe_zero, actual_union_gate_two_summands)
10. *Front total staticized (2026-06-12, 21:17).* For normal-form r (apder_nf) and *every* string s : $\|F(r, s)\| \leq \|\text{apder_rows}(r)\|$ — the dependence on s is gone; $\text{apder_rows}(r)$ is a static, one-regex row universe.
(rsizes_afactored1_le_rsize_set_apder_rows)
11. *Static universe: members quadratic, ledger tightly cubic.* Every member of $\text{apder_rows}(r)$ has size $\leq 1 + (|r| + 2)^2$ ($\text{apder_rows_member_size_quadratic}$); the total ledger is cubic ($\text{rsize_set_apder_rows_expanded_cubic_size_bound}$), and that cubic bound is *tight* (Section 6, item 6).

Current reduction (as of 2026-06-12 23:45, commit 7372ba2). The exact decomposition of the gate object is checked ($\text{actual_union_two_number_decomposition}$):

$$\|U\| = \underbrace{\|\text{nonseq}(U)\|}_{\text{cubic, checked}} + \underbrace{\#\{\text{sequence members}\} \cdot O(|r|)}_{\text{heads linear, checked}} + \sum_{t \in \text{tails}(U)} \text{bucket}(t) \cdot |t|,$$

where $\text{bucket}(t)$ counts the sequence members sharing tail t . By items 10–11 everything is charged to the *static* universe $\text{apder_rows}(r)$, and on the normal-form fragment the conjecture closes by assembly once *either* of two named laws lands:

- (A) *The D law (row-count linearity)*. For normal-form, *legacy* and *ntimes_free* r, k (the guard is load-bearing — see the restriction note below):

$$\#(\text{acc}(r, k) \setminus \text{ftr}(k)) \leq \text{zw}_2(r),$$

where *acc* is the frontier accumulator (*apder_term_frontier_acc*), *ftr* the frontier (*rfrontier*: $\emptyset$ on **0**, union over Σ -members, singleton otherwise), and the weight *zw*₂ (*apder_zw2*) counts letters with a +1 per star: $\text{zw}_2(c) = 1$, sums over $\cdot$ and Σ , $\text{zw}_2(p^*) = 1 + \text{zw}_2(p)$, $\text{zw}_2(p^{\{n\}}) = n \cdot (1 + \text{zw}_2(p))$, zero on **0**, **1**. Since $\text{zw}_2(r) \leq |r|$ on this fragment (*apder_zw2_ntimes_free_le_rsize*, commit 8a7a370), (A) makes $\#\text{apder_rows}(r)$ linear, and members are quadratic (item 11), so the gate ledger is at most linear $\times$ quadratic = cubic, fully statically.

Restriction (do not drop it). The *unrestricted* form of (A) — arbitrary normal-form r, k without the *ntimes_free* guard — is **checked false**: with $r = c \cdot \Sigma[a^{\{0\}}, b^{\{0\}}]$ and $k = \mathbf{1}$ the left side is 2 while $\text{zw}_2(r) = 1$ (*apder_zw2_D_law_ntimes_zero_alt_false*); a zero-count $r^{\{0\}}$ pays no budget yet stays a distinct frontier atom. The live target is the *legacy/ntimes_free* instance only.

Status (2026-06-13 09:18, commit effa163). (A) is no longer attacked as a one-shot bound but by a *simultaneous induction* on a “clean domain” (every $\text{zw}_2=0$ subterm is **0/1**). Landed: the clean-domain passthrough discount (240,256 deep samples, zero violations; STAR branch checked, *discount_RSTAR_if_body_D*); the E00–RONE branch map is *complete* (RCHAR E00_RONE_RCHAR, RALTS E00_RONE_RALTS_if_members, RSTAR-cond. E00_RONE_RSTAR_if_body_D, SEQ-cond. E00_RONE_RSEQ_if); the E0 general- k bridge series is in progress (RCHAR done, E0_RCHAR). Three pieces remain: (a) the union-overlap sublemma (the 4% all-members-passthrough RALTS-discount case where sibling accumulators share imported frontier points); (b) the E0/D general- k branches; (c) the well-founded assembly induction. History to respect: the earlier *zwidth* weight (stars floored at one) and the J^* joint invariant were BOTH falsified by depth-5 counterexamples after passing 200k/95k shallow samples — empirical claims on this problem now require depth ≥ 5 sampling; and *nine* inductive strengthenings are falsified with recorded counterexamples. Self-contained problem file: *MATHPROBLEM_ROWCOUNT.md* (repository root) — do not retry the falsified strengthenings.

- (B) *The liveness/drain law*. The dynamic front stays quadratic, $\|F(r, s)\| \leq C|r|^2$, even on families where the static ledger is genuinely cubic: empirically $0.2\text{--}1.1 \cdot |r|^2$ across all adversarial families with zero violations (only a thin slice of the static rows is simultaneously live at any input position). Reduced to three single-step obligations (cap law); needs a saturation invariant. With (B) the decomposition above makes every gate summand cubic.

Any other route to Conjecture 1 is equally acceptable, but must not pass through the refuted quantities below.

6 What is refuted (do not attempt)

Theorem 1 (duplicated list cost is not polynomial; checked). *Define the list opener D^{list} by the same recursion as D with concatenation instead of union (no deduplication), and the pipeline list cost*

$$\text{listcost}(r, s, c) = \sum_{p \in \text{gen}_c(F(r, s))} \|D^{\text{list}}(S(p))\|.$$

Let $\pi = (\mathbf{1} + a) \cdot (\mathbf{1} + b)$ and the tower $T_0 = k \cdot l$, $T_{n+1} = (\mathbf{1} + \pi) \cdot T_n$ (here $\mathbf{1} + x$ abbreviates $\Sigma[\mathbf{1}, x]$; a, b, k, l characters). Then (all checked):

$$|T_n| = 10n + 3, \quad \text{len } D^{\text{list}}(T_n) = 3 \cdot 2^n - 2, \quad S(T_n) = T_n,$$

and behind one guard character g , $\text{listcost}(g \cdot T_{24}, \varepsilon, g) \geq 3 \cdot 2^{24} - 2 = 50,331,646 > 2(245 + 3)^3 = 30,505,984$, refuting the cubic budget for listcost (`afactored1-strong-dlform-list-cost-cubic-false`).

The mechanism: on $(\Sigma[\mathbf{1}, \pi]) \cdot T$ the list opener recurses into *both* branches; the $\mathbf{1}$ branch collapses to T and reproduces T 's entire opened list, the π branch walks it once more — so the length doubles per level, while the regex grows linearly. The tower is a fixpoint of S , so simplification cannot rescue the cost. Crucially, both branches reach the *same* tail rows: the deduplicated union merges what the list multiplies, which is why Conjecture 1 survives this family (quadratically). Consequences and further checked refutations:

1. Any proof route needing a polynomial bound on listcost, D^{list} -length, or any per-row duplicated multiset is dead (Theorem 1).
2. Generic square-sum wrappers ($\sum_q |q|^2 \leq \|rs\|^2$, or $\text{len} \times \max^3$ products) are true but lose a degree: they yield quartic, not cubic, bounds.
3. The deep-frontier linear-card premise $\#(\text{deep frontier}) \leq \text{awidth} + |r| + 3$ is *false* in general — $r^{\{n\}}$ multiplies alternation branches without paying into the budget (checked witness with cardinality 49 against budget 39; `apder_deep_frontier_linear_card_false`); only the $r^{\{n\}}$ -free version survives. The front-linear variant is likewise refuted (checked, 2026-06-12).
4. Generic owner-closure counting is exponential (checked abstract counterexample, 2026-06-12).
5. S is *not* pointwise D -cost monotone: there is a checked p with $\|D(S(p))\| = 12 > 10 = \|D(p)\|$ (`rsimpStrong_raw_row_dlforms_cost_not_monotone`). Do not assume simplification shrinks the ledger row by row.
6. A *quadratic* static ledger is false: on the star-seq tower $x_0 = a$, $x_{i+1} = (x_i \cdot b_i)^*$ the checked cubic bound `rsizeset_apder_rows_expanded_cubic_size_bound` is tight (ledger $\approx 0.04 |r|^3$ at depth 60; the σ^* star-absorption variant changes nothing on this family). Cubicness of the gate cannot come from shrinking the static universe; it must come from the count side (the D law) or the dynamic slice (liveness).
7. The *unrestricted* (non-`rtimes_free`) `zw2` D law of Section 5, law (A), is *false*: with $r = c \cdot \Sigma[a^{\{0\}}, b^{\{0\}}]$ and $k = \mathbf{1}$ the left side is 2 while $\text{zw2}(r) = 1$ (`apder_zw2_D_law_rtimes_zero_alt_false`). A zero-count $r^{\{0\}}$ contributes no accumulator row and no budget yet remains a distinct frontier atom, so two such alternatives overrun one character's slot. Only the legacy/`rtimes_free` instance is viable; do not re-allow $r^{\{0\}}$ or swap `zw2` for another weight.

7 Acceptance criteria

A solution is a proof of Conjecture 1 checked by Isabelle2025-2 in session `Posix` of the repository (branch `codex/backref-values`), with no `sorry/oops/axioms`, statement guards passing, building via the repository wrapper (`scripts/codex-isabelle-build-posix.ps1`). Per the 2026-06-12 bounty overlay (simulated USD): 12,000 for the final checked theorem (or a checked equivalent contract) preserving exact POSIX value behaviour; 5,000 for a major checked bridge that reduces it to a clearly smaller, non-false local obligation (laws (A)/(B) above are the

current such targets); 3,000 for a checked negative result that materially redirects the search. Fragment results ($r^{\{n\}}$ -free) and constant-relaxed versions count as bridges, per Remark 1.

Live status, checked-facts table and dead-route list: `MAINLINE.md`; document catalog: `DOC_INDEX.md`; day-by-day state: tail of `PROGRESS_BACKREF.md`; the D law as a standalone problem: `MATHPROBLEM.ROWCOUNT.md` (all at the repository root).

8 Notation–Isabelle dictionary

here	Isabelle name	file
$ r $, $\ rs\ $	<code>rsiz</code> , <code>rsizes</code>	BI
$\ U\ $ (set)	<code>rsiz_set</code>	AFT
flts , nub , $\oplus$	<code>rflts</code> , <code>rdistinct</code> , <code>rsimp_ALT</code> s	BI
σ , σ^*	<code>rsimp4_SEQ_atom</code> , <code>rsimp7_SEQ_atom</code>	BI
pder_c , npder_c	<code>rpder_list</code> , <code>rpder_norm_list</code>	GRB
step_c , F	<code>afactored_step</code> , <code>afactored1</code>	AFT
P , prune	<code>rsimpStrong_prune_pair_raw</code> , <code>..._prune_rows_raw</code>	GRB
S	<code>rsimpStrong_raw</code>	GRB
Δ_c	<code>rpder_strong_rows_raw</code>	GRB
D	<code>row_dlforms</code> , <code>row_dlformss</code>	AFT
D^{list} , listcost	<code>row_dlforms_list</code> , <code>afactored1_strong_dlform_list_cost</code>	AFT
fronts , pairbudget	<code>strong_derivative_front_terms</code> , <code>raw_shared_prune_active_suffix_pair_budget</code>	AFT / GRB
apder_rows , ftr	<code>apder_rows</code> , <code>rfrontier</code>	AFT / GRB
acc , zw_2	<code>apder_term_frontier_acc</code> , <code>apder_zw2</code>	AFT
target gate	<code>assumption actual_cubic of ..._tight_cubic_contractl</code>	AFT

AFT = `AntimirovFactoredTransition.thy`; GRB = `GeneralRegexBound.thy`; BI = `BasicIdentities.thy`.